XUAN-NGHIA HUYNH

PhD Candidate

hxnghia99k@gmail.com | 010-9985-2279 | Seoul, Korea | linkedin.com/in/x-nghia-huynh | github.com/hxnghia99

Summary

Innovative AI PhD Candidate with expertise in Computer Vision (Object Detection and Segmentation) and Generative AI (GANs, Diffusion Models). Specialized in mounted-camera detection applications, layout-conditioned and multi-spectral image synthesis with automated annotation. Seeking internship or full-time roles in AI Research, Computer Vision Engineering, or Generative AI Development.

Education

PhD Candidate in AI & Robotics, *Sejong University* — 03/2022 – Present
Expected Graduation: Feb 2027 — Seoul, South Korea
Dissertation (ongoing): Generative Frameworks for Layout-Conditioned RGB and Multi-Spectral Fire Image Synthesis to Enhance Detection and Segmentation Tasks.
GPA: 4.41/4.5

Bachelor of Electrical and Electronics Engineering, — 09/2017 – 08/2021
Vietnam National University Ho Chi Minh City - Ho Chi Minh City University of Technology (HCMUT) — HCM City, Vietnam
Thesis: Mobile Robot Applied 3D SLAM & Deep Learning
GPA: 8.62/10
Member of Honor Class of Control Engineering and Automation

Professional Experience

PhD Researcher, *IVPG Laboratory, Sejong University* — 03/2022 – Present
Project 1: Small Object Detection for Real-Time Embedded System (First Author) — Seoul, South Korea

- **Designed and implemented** a lightweight Super-Resolution module (SRm) for YOLOv4 in TensorFlow to enhance small object detection in surveillance and drone imagery while minimizing computational cost.
- **Achieved** YOLOv4-SRm model with a lower number of parameters compared to vanilla YOLOv4, while increasing AP50 on the company dataset and VisDrone2019 dataset by 11.38% and 6.29%, respectively.
- **Developed** a simplified SRm (SSRm) variant for YOLOv4 architecture with a lower number of parameters, similar FLOPs, and higher AP50 compared to vanilla YOLOv4. **Cooperated** to conduct pruning and quantization on the YOLOv4-SSRm model, and embedded it on Qualcomm QCS610 SoC using Qualcomm® Neural Processing SDK (SNPE), achieving detection speed at 36 FPS on the company dataset, contributing to a real-time CCTV detection framework.
- **Contributed** to the research project by performing literature review and data analysis, conducting experiments and comparisons, supporting embedded implementation, and writing manuscripts.

Project 2: GAN-based Layout-Guided Fire Image & Mask Generation (First Author)

- **Proposed and built** an end-to-end layout-conditioned GAN-based approach on PyTorch framework utilizing a combination of Image-to-Image and Layout-to-Image translations to simultaneously generate photorealistic fire images and segmentation masks, lowering FID score by 4.79 compared to the PaintByExample diffusion model.
- **Accomplished** a fire data augmentation pipeline that produced synthesized fire images along with bounding box annotations using COCO format and semantic segmentation masks of fire objects, without manual annotation effort.
- **Validated** data augmentation efficiency on a self-built fire dataset using YOLOv9 for detection task (increased mAP by 4.6%) and a customized U-Net model for segmentation task (enhanced mIoU by 4.44%)
- **Contributed** to the research project by performing literature review and data analysis, conducting experiments and comparisons, and writing manuscripts.

Project 3: Diffusion-based Layout-Conditioned Multi-Spectral Fire Synthesis (First Author) (ongoing)

- **Proposed and developed** a layout-conditioned Diffusion Model on PyTorch framework for RGB-only fire image synthesis by combining concepts of Image-to-Image and Layout-to-Image translations, using a layout-aware cross-attention mechanism, and validated by successfully generating fire images with two classes (fire & smoke) from non-fire backgrounds.
- **Developed** a layout-conditioned Diffusion Model for synchronized cross-spectral (RGB & Near-Infrared / NIR) fire image synthesis using channel concatenation and background conditioning with a layout-aware cross-attention mechanism, simultaneously producing synchronized outputs: RGB and NIR fire images.
- **Formulated** a cross-attention conditioning mechanism for automatic segmentation mask refinement during reverse process, reducing manual annotation effort.

- **Contributed** to the research project by performing literature review and data analysis, conducting experiments and comparisons, and writing manuscripts (ongoing)

Undergraduate Research Assistant, *Autonomic Control Lab, HCMUT* 01/2020 – 08/2021
Project: Mobile Robot Applied 3D SLAM & Deep Learning (Co-author) HCM City, Vietnam

- **Configured and simulated** a mobile robot applying 2D Hector SLAM algorithm in Gazebo tool, visualized the generated 2D map, and conducted navigation tasks of this robot in RViz tool, indicating practical feasibility.
- **Designed and built** a three-wheel mobile robot using SolidWorks, **programmed and built** ROS1 packages using RTABMAP algorithm (a 3D SLAM method) on Ubuntu platform using RPLidar and RealSense D435i Camera. **Utilized and built** the localization task using AMCL filter, the navigation task using Dijkstra's algorithm, and obstacle avoidance using Dynamic Window algorithm.
- **Cooperated** in integrating 3D SLAM with deep-learning-based human tracking algorithm by server-client architecture via Wi-Fi protocol, achieving the task of detection and tracking.

Embedded Software Engineer (Intern), *Robert Bosch Engineering and Business Solutions VN* 06/2020 – 12/2020
Worked as a software developer for RADAR team HCM City, Vietnam

- Worked and researched AUTOSAR architecture in the communication application layer.
- Studied and researched CAN protocol in automotive embedded systems.

Skills

Generative AI
Diffusion Models (DDPM, DDIM, Latent Diffusion), GANs (CycleGAN, SPADEGAN, LostGAN), Cross-spectral Image Translation, Transformer Attention, Layout-conditioning

Computer Vision
Object Detection, Semantic Segmentation, YOLO (v4, v9), Knowledge Distillation, Multi-GPU training, Deep Learning, CNN, Classification

Programming Languages & Frameworks
Python (PyTorch, TorchVision, OpenCV, Pandas, NumPy, TensorFlow), C/C++

Tools
Git, Docker, Linux, Jupyter Notebook, HuggingFace, Microsoft Azure

Soft Skills
Analytical thinking, Problem-solving, Creativity, Adaptability

Languages
Vietnamese (Native), English (Fluent, IELTS 7.0), Korean (Basic)

Publications

X.N. Huynh, and J.K. Suhr, "Fire Image Generation Based on Image-to-Image and Layout-to-Image Translations for Detection and Segmentation Tasks", *IEEE Access* 2026

X.N. Huynh, and J.K. Suhr, "Diffusion-based Fire Image Synthesis for Generating Object Detection Datasets", *Proceedings of the 2025 Summer Conference of The Institute of Electronics and Information Engineers* 2025

X.N. Huynh, G.B. Jung, and J.K. Suhr, "One-Stage Small Object Detection Using Super-Resolved Feature Map for Edge Devices", *MDPI Electronics* 2024

References

Jae Kyu Suhr, *Associate Professor,* Sejong University
jksuhr@sejong.ac.kr